D4.3 Climate change

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d4-3.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 834–837 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Timing

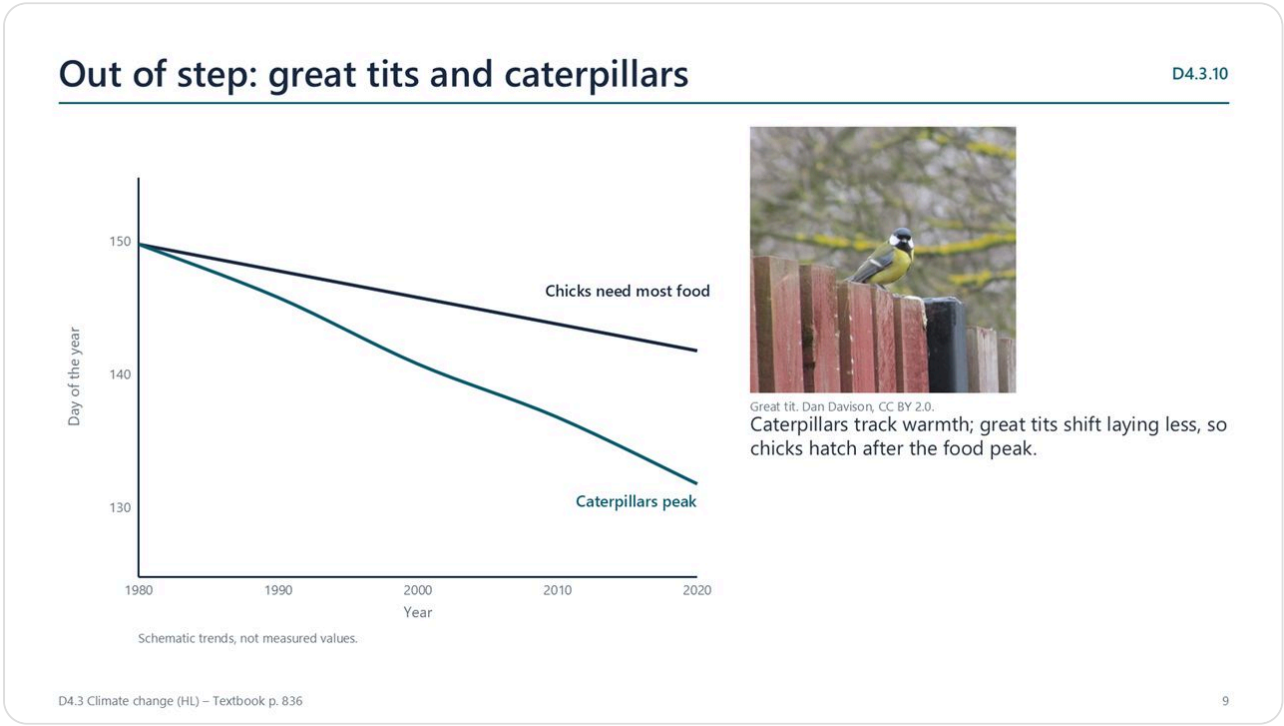
D4.3.9 Phenology

p. 834

Phenology is the study of the timing of biological events such as flowering, budburst, bud set, migration and nesting. Day length and temperature are the main cues.

D4.3.10 Loss of synchrony

pp. 835–836



Species may use different cues, so warming can put linked events out of step. Arctic plants grow earlier while reindeer still arrive by day length. Caterpillars peak earlier, and great tit chicks can hatch too late for them.

Life cycles and evolution

D4.3.11 More insect generations

pp. 836–837

Warmth speeds insect development. Spruce bark beetles in northern regions can move from one generation a year to two, so more beetles attack each tree and more trees die.

D4.3.12 Evolution

p. 837

With less snow, pale tawny owls are more visible and catch less food, so brown owls have become more common in Finland. In Scots pine, a longer growing season favours later bud set, so alleles for later bud set can rise in frequency.

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